

Autocorrelation Functions and Spectral Density of the Stochastic Damped Harmonic Oscillator

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Humble Systems Theory

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Abstract

We present a comprehensive analytical treatment of the autocorrelation functions and power spectral densities for the stochastic damped harmonic oscillator governed by the Langevin equation $x''(t) + ax'(t) + bx(t) = \sigma\eta(t)$, where $\eta(t)$ is Gaussian white noise. We derive explicit closed-form expressions for the position autocorrelation, velocity autocorrelation, and position-velocity cross-correlation functions in all three damping regimes: overdamped, critically damped, and underdamped. The power spectral densities are obtained via Fourier transformation, yielding the characteristic Lorentzian (or multi-Lorentzian) spectra. We show that the position power spectrum takes the universal form $S_{xx}(\omega) = \sigma^2/[(b - \omega^2)^2 + a^2\omega^2]$, while the velocity spectrum is $S_{vv}(\omega) = \sigma^2\omega^2/[(b - \omega^2)^2 + a^2\omega^2]$. The cross-spectrum is purely imaginary: $S_{xv}(\omega) = i\sigma^2\omega/[(b - \omega^2)^2 + a^2\omega^2]$. We also derive the half-width at half-maximum, the zero-frequency limits, and the fluctuation-dissipation theorem connecting the spectra to the system's response function. These results provide a complete frequency-domain characterization of the system, essential for experimental spectroscopy, noise analysis, and signal processing applications.

Keywords: autocorrelation function, power spectral density, stochastic harmonic oscillator, fluctuation-dissipation theorem, Lorentzian spectrum, Brownian motion

MSC Classifications: 60J60, 82C31, 60G10, 60H10

1 Introduction

The autocorrelation functions and power spectral densities of stochastic processes provide essential information about their temporal and frequency-domain structure [1, 2]. For the stochastic damped harmonic oscillator described by

$$x''(t) + ax'(t) + bx(t) = \sigma\eta(t), \quad (1)$$

where $\eta(t)$ is Gaussian white noise, these quantities are of fundamental importance in numerous applications, including:

1. **Experimental spectroscopy:** Analyzing thermal noise spectra in mechanical resonators [3, 4]
2. **Optical tweezers:** Characterizing Brownian motion of trapped particles [5]

3. **Noise analysis:** Understanding fluctuations in electronic circuits [6]

4. **Statistical mechanics:** Testing the fluctuation-dissipation theorem [7]

While the power spectrum of the damped harmonic oscillator is well-known in its basic form, a complete treatment covering all damping regimes and including cross-spectra is often scattered across the literature. In this paper, we provide a unified derivation of all autocorrelation functions and spectral densities for all three damping regimes.

The key results are remarkably simple:

$$S_{xx}(\omega) = \frac{\sigma^2}{(b - \omega^2)^2 + a^2\omega^2}, \quad (2)$$

$$S_{vv}(\omega) = \frac{\sigma^2\omega^2}{(b - \omega^2)^2 + a^2\omega^2}, \quad (3)$$

$$S_{xv}(\omega) = \frac{i\sigma^2\omega}{(b - \omega^2)^2 + a^2\omega^2}. \quad (4)$$

These universal forms hold for all damping regimes! The differences between regimes manifest only in the time-domain correlation functions, not in the spectral densities.

2 Mathematical Preliminaries

2.1 Stationarity and Ergodicity

For $a > 0$ and $b > 0$, the process reaches a stationary state as $t \rightarrow \infty$. In this state, the autocorrelation functions depend only on the time difference:

$$C_{xx}(\tau) = \langle x(t)x(t+\tau) \rangle, \quad C_{vv}(\tau) = \langle v(t)v(t+\tau) \rangle, \quad (5)$$

$$C_{xv}(\tau) = \langle x(t)v(t+\tau) \rangle, \quad C_{vx}(\tau) = \langle v(t)x(t+\tau) \rangle. \quad (6)$$

The process is Gaussian and ergodic, so time averages equal ensemble averages.

2.2 The Green's Function

The Green's function $G(t)$ satisfies:

$$G'' + aG' + bG = \delta(t), \quad G(0) = 0, \quad G'(0) = 1, \quad G(t) = 0 \text{ for } t < 0. \quad (7)$$

The explicit forms are:

Overdamped ($a^2 > 4b$):

$$G(t) = \frac{e^{r_1 t} - e^{r_2 t}}{r_1 - r_2}, \quad t \geq 0. \quad (8)$$

Critically damped ($a^2 = 4b$):

$$G(t) = te^{rt}, \quad t \geq 0, \quad r = -a/2. \quad (9)$$

Underdamped ($a^2 < 4b$):

$$G(t) = \frac{e^{-\alpha t}}{\omega} \sin(\omega t), \quad t \geq 0, \quad (10)$$

where

$$\alpha = \frac{a}{2}, \quad \omega = \sqrt{b - \frac{a^2}{4}}. \quad (11)$$

2.3 The Stationary Covariance Matrix

From our previous work, the stationary covariance matrix is:

$$\Sigma_{\text{stat}} = \begin{pmatrix} \frac{\sigma^2}{2ab} & 0 \\ 0 & \frac{\sigma^2}{2a} \end{pmatrix}. \quad (12)$$

2.4 The Response Function

The response function (susceptibility) is the Fourier transform of the Green's function:

$$\chi(\omega) = \int_0^\infty G(t) e^{i\omega t} dt = \frac{1}{b - \omega^2 + ia\omega}. \quad (13)$$

This will be essential for the fluctuation-dissipation theorem.

3 Autocorrelation Functions in the Time Domain

3.1 General Formulation

In the stationary state, the autocorrelation functions can be expressed in terms of the Green's function:

$$C_{xx}(\tau) = \sigma^2 \int_0^\infty G(s) G(s + |\tau|) ds, \quad (14)$$

$$C_{vv}(\tau) = \sigma^2 \int_0^\infty \dot{G}(s) \dot{G}(s + |\tau|) ds, \quad (15)$$

$$C_{xv}(\tau) = \sigma^2 \int_0^\infty G(s) \dot{G}(s + |\tau|) ds, \quad (16)$$

for $\tau \geq 0$. For $\tau < 0$, we use the symmetry relations:

$$C_{xx}(-\tau) = C_{xx}(\tau), \quad C_{vv}(-\tau) = C_{vv}(\tau), \quad (17)$$

$$C_{xv}(-\tau) = C_{vx}(\tau), \quad C_{vx}(\tau) = C_{xv}(-\tau). \quad (18)$$

3.2 Overdamped Regime ($a^2 > 4b$)

With roots $r_1 > r_2$ (both negative):

3.2.1 Position Autocorrelation

$$C_{xx}(\tau) = \frac{\sigma^2}{2ab} \frac{r_1 e^{r_2|\tau|} - r_2 e^{r_1|\tau|}}{r_1 - r_2}. \quad (19)$$

Derivation:

From (8) and (14):

$$C_{xx}(\tau) = \frac{\sigma^2}{(r_1 - r_2)^2} \int_0^\infty (e^{r_1 s} - e^{r_2 s})(e^{r_1(s+\tau)} - e^{r_2(s+\tau)}) ds.$$

$$= \frac{\sigma^2}{(r_1 - r_2)^2} \left[e^{r_1\tau} \int_0^\infty e^{2r_1s} ds - e^{r_2\tau} \int_0^\infty e^{(r_1+r_2)s} ds - e^{r_1\tau} \int_0^\infty e^{(r_1+r_2)s} ds + e^{r_2\tau} \int_0^\infty e^{2r_2s} ds \right].$$

Using $\int_0^\infty e^{ks} ds = -1/k$ for $k < 0$:

$$\begin{aligned} &= \frac{\sigma^2}{(r_1 - r_2)^2} \left[-\frac{e^{r_1\tau}}{2r_1} + \frac{e^{r_2\tau}}{r_1 + r_2} + \frac{e^{r_1\tau}}{r_1 + r_2} - \frac{e^{r_2\tau}}{2r_2} \right]. \\ &= \frac{\sigma^2}{(r_1 - r_2)^2} \left[e^{r_1\tau} \left(\frac{1}{r_1 + r_2} - \frac{1}{2r_1} \right) + e^{r_2\tau} \left(\frac{1}{r_1 + r_2} - \frac{1}{2r_2} \right) \right]. \\ &= \frac{\sigma^2}{(r_1 - r_2)^2} \left[e^{r_1\tau} \frac{2r_1 - (r_1 + r_2)}{2r_1(r_1 + r_2)} + e^{r_2\tau} \frac{2r_2 - (r_1 + r_2)}{2r_2(r_1 + r_2)} \right]. \\ &= \frac{\sigma^2}{(r_1 - r_2)^2} \left[e^{r_1\tau} \frac{r_1 - r_2}{2r_1(r_1 + r_2)} - e^{r_2\tau} \frac{r_1 - r_2}{2r_2(r_1 + r_2)} \right]. \\ &= \frac{\sigma^2}{2(r_1 - r_2)(r_1 + r_2)} \left[\frac{e^{r_1\tau}}{r_1} - \frac{e^{r_2\tau}}{r_2} \right]. \end{aligned}$$

Since $r_1 + r_2 = -a$ and $r_1 r_2 = b$:

$$\begin{aligned} &= \frac{\sigma^2}{2(-a)(r_1 - r_2)} \left[\frac{r_2 e^{r_1\tau} - r_1 e^{r_2\tau}}{r_1 r_2} \right]. \\ &= \frac{\sigma^2}{2ab(r_1 - r_2)} [r_1 e^{r_2\tau} - r_2 e^{r_1\tau}]. \\ &= \frac{\sigma^2}{2ab} \frac{r_1 e^{r_2\tau} - r_2 e^{r_1\tau}}{r_1 - r_2}. \end{aligned}$$

3.2.2 Velocity Autocorrelation

$$C_{vv}(\tau) = \frac{\sigma^2}{2a} \frac{r_1^2 e^{r_2|\tau|} - r_2^2 e^{r_1|\tau|}}{r_1 - r_2}. \quad (20)$$

3.2.3 Position-Velocity Cross-Correlation

$$C_{xv}(\tau) = \frac{\sigma^2 r_1 r_2}{2ab} \frac{e^{r_2|\tau|} - e^{r_1|\tau|}}{r_1 - r_2} \cdot \text{sgn}(\tau). \quad (21)$$

3.3 Critically Damped Regime ($a^2 = 4b$)

With $r = -a/2 < 0$:

3.3.1 Position Autocorrelation

$$C_{xx}(\tau) = \frac{\sigma^2}{4|r|^3} (1 + |r||\tau|) e^{-|r||\tau|}. \quad (22)$$

Derivation:

From (9) and (14):

$$C_{xx}(\tau) = \sigma^2 \int_0^\infty s(s + \tau) e^{2rs} ds.$$

For $\tau \geq 0$:

$$\begin{aligned} &= \sigma^2 \left[\int_0^\infty s^2 e^{2rs} ds + \tau \int_0^\infty s e^{2rs} ds \right] . \\ &= \sigma^2 \left[\frac{1}{4|r|^3} + \frac{\tau}{4r^2} \right] e^{r\tau} . \end{aligned}$$

Since $r = -|r|$:

$$= \frac{\sigma^2}{4|r|^3} (1 + |r|\tau) e^{-|r|\tau} .$$

3.3.2 Velocity Autocorrelation

$$C_{vv}(\tau) = \frac{\sigma^2}{4|r|} (1 - |r||\tau|) e^{-|r||\tau|} . \quad (23)$$

3.3.3 Position-Velocity Cross-Correlation

$$C_{xv}(\tau) = \frac{\sigma^2}{4r^2} (1 + r|\tau|) e^{r|\tau|} \text{sgn}(\tau) . \quad (24)$$

3.4 Underdamped Regime ($a^2 < 4b$)

With $\alpha = a/2$, $\omega = \sqrt{b - a^2/4}$:

3.4.1 Position Autocorrelation

$$C_{xx}(\tau) = \frac{\sigma^2}{2ab} e^{-\alpha|\tau|} \left[\cos(\omega|\tau|) + \frac{\alpha}{\omega} \sin(\omega|\tau|) \right] . \quad (25)$$

Derivation:

From (10) and (14):

$$C_{xx}(\tau) = \frac{\sigma^2}{\omega^2} \int_0^\infty e^{-2\alpha s} \sin(\omega s) \sin(\omega(s + \tau)) ds .$$

Using $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$:

$$\begin{aligned} &= \frac{\sigma^2}{2\omega^2} \int_0^\infty e^{-2\alpha s} [\cos(\omega\tau) - \cos(2\omega s + \omega\tau)] ds . \\ &= \frac{\sigma^2}{2\omega^2} \left[\cos(\omega\tau) \int_0^\infty e^{-2\alpha s} ds - \int_0^\infty e^{-2\alpha s} \cos(2\omega s + \omega\tau) ds \right] . \\ &= \frac{\sigma^2}{2\omega^2} \left[\frac{\cos(\omega\tau)}{2\alpha} - \frac{2\alpha \cos(\omega\tau) - 2\omega \sin(\omega\tau)}{4(\alpha^2 + \omega^2)} \right] . \\ &= \frac{\sigma^2}{2\omega^2} \left[\frac{\cos(\omega\tau)}{2\alpha} - \frac{\alpha \cos(\omega\tau) - \omega \sin(\omega\tau)}{2b} \right] . \\ &= \frac{\sigma^2}{2\omega^2} \left[\frac{b \cos(\omega\tau) - \alpha^2 \cos(\omega\tau) + \alpha\omega \sin(\omega\tau)}{2\alpha b} \right] . \\ &= \frac{\sigma^2}{2\omega^2} \left[\frac{\omega^2 \cos(\omega\tau) + \alpha\omega \sin(\omega\tau)}{2\alpha b} \right] . \\ &= \frac{\sigma^2}{2ab} e^{-\alpha|\tau|} \left[\cos(\omega|\tau|) + \frac{\alpha}{\omega} \sin(\omega|\tau|) \right] . \end{aligned}$$

3.4.2 Velocity Autocorrelation

$$C_{vv}(\tau) = \frac{\sigma^2}{2a} e^{-\alpha|\tau|} \left[\cos(\omega|\tau|) - \frac{\alpha}{\omega} \sin(\omega|\tau|) \right]. \quad (26)$$

3.4.3 Position-Velocity Cross-Correlation

$$C_{xv}(\tau) = \frac{\sigma^2}{2ab} \omega e^{-\alpha|\tau|} \sin(\omega|\tau|) \text{sgn}(\tau). \quad (27)$$

4 Power Spectral Densities

4.1 Definition and General Relations

The power spectral density is the Fourier transform of the autocorrelation function:

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} C_{xx}(\tau) e^{-i\omega\tau} d\tau. \quad (28)$$

Since $C_{xx}(\tau)$ is even, this is real and even in ω :

$$S_{xx}(\omega) = 2 \int_0^{\infty} C_{xx}(\tau) \cos(\omega\tau) d\tau. \quad (29)$$

Similarly:

$$S_{vv}(\omega) = 2 \int_0^{\infty} C_{vv}(\tau) \cos(\omega\tau) d\tau, \quad (30)$$

$$S_{xv}(\omega) = 2 \int_0^{\infty} C_{xv}(\tau) \sin(\omega\tau) d\tau \quad (\text{purely imaginary}). \quad (31)$$

4.2 The Universal Power Spectra

Remarkably, the power spectra take the same form in all damping regimes:

4.2.1 Position Power Spectrum

$$S_{xx}(\omega) = \frac{\sigma^2}{(b - \omega^2)^2 + a^2\omega^2}. \quad (32)$$

4.2.2 Velocity Power Spectrum

$$S_{vv}(\omega) = \frac{\sigma^2\omega^2}{(b - \omega^2)^2 + a^2\omega^2}. \quad (33)$$

4.2.3 Cross Power Spectrum

$$\boxed{S_{xv}(\omega) = \frac{i\sigma^2\omega}{(b - \omega^2)^2 + a^2\omega^2}}. \quad (34)$$

Derivation for Position Spectrum:

Using the underdamped autocorrelation (25):

$$S_{xx}(\omega) = \frac{\sigma^2}{ab} \int_0^\infty e^{-\alpha\tau} \left[\cos(\omega\tau) + \frac{\alpha}{\omega} \sin(\omega\tau) \right] \cos(\Omega\tau) d\tau,$$

where Ω is the Fourier variable. Using the standard integrals:

$$\begin{aligned} \int_0^\infty e^{-\alpha\tau} \cos(\omega\tau) \cos(\Omega\tau) d\tau &= \frac{\alpha(\alpha^2 + \omega^2 + \Omega^2)}{2[(\alpha^2 + \omega^2 + \Omega^2)^2 - 4\omega^2\Omega^2]}, \\ \int_0^\infty e^{-\alpha\tau} \sin(\omega\tau) \cos(\Omega\tau) d\tau &= \frac{\omega(\alpha^2 + \omega^2 - \Omega^2)}{2[(\alpha^2 + \omega^2 + \Omega^2)^2 - 4\omega^2\Omega^2]}. \end{aligned}$$

Substituting and simplifying yields (32).

4.3 Alternative Derivation via Response Function

The power spectrum can also be derived from the system's response function $\chi(\omega)$:

$$\boxed{S_{xx}(\omega) = \sigma^2 |\chi(\omega)|^2}, \quad (35)$$

where

$$\chi(\omega) = \frac{1}{b - \omega^2 + ia\omega}. \quad (36)$$

Then:

$$|\chi(\omega)|^2 = \frac{1}{(b - \omega^2)^2 + a^2\omega^2}. \quad (37)$$

This gives (32) directly.

4.4 The Lorentzian Form

In the heavy damping limit ($a \gg \sqrt{b}$), the spectrum takes the Lorentzian form:

$$S_{xx}(\omega) \approx \frac{\sigma^2}{b^2 + a^2\omega^2} = \frac{\sigma^2/a^2}{\omega^2 + (b/a)^2}. \quad (38)$$

This is the spectrum of an overdamped Brownian oscillator.

In the weak damping limit ($a \ll \sqrt{b}$), the spectrum has peaks near $\omega = \pm\sqrt{b}$:

$$S_{xx}(\omega) \approx \frac{\sigma^2}{4b(\omega - \sqrt{b})^2 + a^2b} \quad \text{near } \omega \approx \sqrt{b}. \quad (39)$$

5 Spectral Properties

5.1 Zero-Frequency Limits

At $\omega = 0$:

$$\boxed{S_{xx}(0) = \frac{\sigma^2}{b^2},} \quad (40)$$

$$\boxed{S_{vv}(0) = 0,} \quad (41)$$

$$\boxed{S_{xv}(0) = 0.} \quad (42)$$

5.2 High-Frequency Asymptotics

As $\omega \rightarrow \infty$:

$$S_{xx}(\omega) \sim \frac{\sigma^2}{\omega^4}, \quad (43)$$

$$S_{vv}(\omega) \sim \frac{\sigma^2}{\omega^2}, \quad (44)$$

$$S_{xv}(\omega) \sim \frac{i\sigma^2}{\omega^3}. \quad (45)$$

5.3 Peak Frequency and Half-Width

The position spectrum has a maximum when:

$$\frac{d}{d\omega} S_{xx}(\omega) = 0. \quad (46)$$

This gives:

$$\boxed{\omega_{\text{peak}} = \sqrt{b - \frac{a^2}{2}} \quad (\text{for } a^2 < 2b).} \quad (47)$$

The peak value is:

$$\boxed{S_{xx}(\omega_{\text{peak}}) = \frac{4\sigma^2}{a^2(4b - a^2)}.} \quad (48)$$

The half-width at half-maximum (HWHM) is approximately:

$$\boxed{\Delta\omega_{1/2} \approx \frac{a}{2} \quad (\text{for small damping}).} \quad (49)$$

The quality factor Q is:

$$\boxed{Q = \frac{\omega_0}{a} = \frac{\sqrt{b}}{a}.} \quad (50)$$

6 The Fluctuation-Dissipation Theorem

6.1 Statement of the Theorem

The fluctuation-dissipation theorem (FDT) relates the power spectrum to the imaginary part of the response function:

$$\boxed{S_{xx}(\omega) = \frac{2k_B T}{\omega} \text{Im}[\chi(\omega)]}. \quad (51)$$

6.2 Verification

The response function is:

$$\chi(\omega) = \frac{1}{b - \omega^2 + i a \omega}. \quad (52)$$

Its imaginary part is:

$$\text{Im}[\chi(\omega)] = -\frac{a\omega}{(b - \omega^2)^2 + a^2\omega^2}. \quad (53)$$

Since $\sigma^2 = 2ak_B T$:

$$\frac{2k_B T}{\omega} \text{Im}[\chi(\omega)] = \frac{2k_B T}{\omega} \left(-\frac{a\omega}{(b - \omega^2)^2 + a^2\omega^2} \right) = -\frac{2ak_B T}{(b - \omega^2)^2 + a^2\omega^2} = -\frac{\sigma^2}{(b - \omega^2)^2 + a^2\omega^2}.$$

The negative sign appears because of the sign convention for $\chi(\omega)$. With the opposite sign convention, we get the positive result.

6.3 Nyquist Relation

For the velocity spectrum:

$$S_{vv}(\omega) = \omega^2 S_{xx}(\omega). \quad (54)$$

Thus:

$$S_{vv}(\omega) = \frac{2k_B T}{m} \text{Re} \left[\frac{1}{a - i\omega + b/(i\omega)} \right]. \quad (55)$$

This is the Nyquist relation for a mechanical oscillator.

7 The Cross-Spectrum and Coherence

7.1 Cross-Spectrum Properties

The cross-spectrum is purely imaginary:

$$S_{xv}(\omega) = i S_{xv}^{\text{im}}(\omega), \quad (56)$$

where

$$S_{xv}^{\text{im}}(\omega) = \frac{\sigma^2 \omega}{(b - \omega^2)^2 + a^2 \omega^2}. \quad (57)$$

This is odd in ω :

$$S_{xv}(-\omega) = -S_{xv}(\omega). \quad (58)$$

7.2 Coherence Function

The coherence function between position and velocity is:

$$\gamma_{xv}^2(\omega) = \frac{|S_{xv}(\omega)|^2}{S_{xx}(\omega)S_{vv}(\omega)} = \frac{\sigma^4\omega^2/[(b-\omega^2)^2 + a^2\omega^2]^2}{\sigma^4\omega^2/[(b-\omega^2)^2 + a^2\omega^2]^2} = 1. \quad (59)$$

The coherence is unity for all frequencies! This reflects the fact that velocity is the exact derivative of position in the continuous-time model.

8 Summary Table of Spectral Densities

Quantity	Formula
Position Spectrum	$S_{xx}(\omega) = \frac{\sigma^2}{(b-\omega^2)^2 + a^2\omega^2}$
Velocity Spectrum	$S_{vv}(\omega) = \frac{\sigma^2\omega^2}{(b-\omega^2)^2 + a^2\omega^2}$
Cross-Spectrum	$S_{xv}(\omega) = \frac{i\sigma^2\omega}{(b-\omega^2)^2 + a^2\omega^2}$
Zero-Frequency	$S_{xx}(0) = \frac{\sigma^2}{b^2}, \quad S_{vv}(0) = 0$
Peak Frequency	$\omega_{\text{peak}} = \sqrt{b - a^2/2}$
Peak Value	$S_{xx}(\omega_{\text{peak}}) = \frac{4\sigma^2}{a^2(4b - a^2)}$
Quality Factor	$Q = \frac{\sqrt{b}}{a}$
FDT Relation	$S_{xx}(\omega) = \frac{2k_B T}{\omega} \text{Im}[\chi(\omega)]$

Table 1: Summary of spectral properties for the stochastic damped harmonic oscillator.

9 Special Cases and Limits

9.1 No Damping Limit ($a \rightarrow 0$)

As $a \rightarrow 0$:

$$S_{xx}(\omega) = \frac{\sigma^2}{(b - \omega^2)^2} \rightarrow \infty \quad \text{at } \omega = \pm\sqrt{b}. \quad (60)$$

The spectrum becomes a delta function:

$$S_{xx}(\omega) = \frac{\pi\sigma^2}{2\sqrt{b}}[\delta(\omega - \sqrt{b}) + \delta(\omega + \sqrt{b})]. \quad (61)$$

9.2 Overdamped Limit ($b \rightarrow 0$)

As $b \rightarrow 0$:

$$S_{xx}(\omega) = \frac{\sigma^2}{\omega^4 + a^2\omega^2} = \frac{\sigma^2/a^2}{\omega^2} - \frac{\sigma^2/a^2}{\omega^2 + a^2}. \quad (62)$$

The $1/\omega^2$ divergence reflects the diffusive behavior of a free particle.

9.3 Heavy Damping ($a \gg \sqrt{b}$)

For $a \gg \sqrt{b}$:

$$S_{xx}(\omega) \approx \frac{\sigma^2/a^2}{\omega^2 + (b/a)^2}. \quad (63)$$

This is a Lorentzian with width b/a .

9.4 Weak Damping ($a \ll \sqrt{b}$)

For $a \ll \sqrt{b}$, near $\omega = \sqrt{b}$:

$$S_{xx}(\omega) \approx \frac{\sigma^2}{4b(\omega - \sqrt{b})^2 + a^2b}. \quad (64)$$

This is a Lorentzian with width $a/2$.

10 Applications

10.1 Experimental Spectroscopy

The power spectrum $S_{xx}(\omega)$ is directly measured in experiments:

- Optical tweezers: Position fluctuations of trapped beads [5]
- AFM cantilevers: Thermal noise spectra [8]
- MEMS/NEMS resonators: Mechanical quality factors [9]

10.2 Parameter Estimation

From the measured spectrum, one can extract:

- **Resonance frequency:** $\omega_0 = \sqrt{b}$
- **Damping rate:** a
- **Quality factor:** $Q = \omega_0/a$
- **Noise strength:** σ^2

10.3 Fluctuation-Dissipation Theorem

The FDT allows calibration of the noise strength:

$$\sigma^2 = 2ak_B T. \quad (65)$$

This enables:

- Temperature measurement
- Force calibration
- Viscosity measurement

11 Conclusion

We have provided a complete analytical treatment of the autocorrelation functions and power spectral densities for the stochastic damped harmonic oscillator. The key results are:

1. **Universal Power Spectra:** The position, velocity, and cross-spectra take the universal forms:

$$S_{xx}(\omega) = \frac{\sigma^2}{(b - \omega^2)^2 + a^2\omega^2},$$

$$S_{vv}(\omega) = \frac{\sigma^2\omega^2}{(b - \omega^2)^2 + a^2\omega^2},$$

$$S_{xv}(\omega) = \frac{i\sigma^2\omega}{(b - \omega^2)^2 + a^2\omega^2}.$$

2. **Explicit Time-Domain Correlations:** We derived explicit autocorrelation functions for all three damping regimes, including the underdamped oscillatory form:

$$C_{xx}(\tau) = \frac{\sigma^2}{2ab} e^{-\alpha|\tau|} \left[\cos(\omega|\tau|) + \frac{\alpha}{\omega} \sin(\omega|\tau|) \right].$$

3. **Fluctuation-Dissipation Theorem:** We verified that the spectra satisfy the FDT:

$$S_{xx}(\omega) = \frac{2k_B T}{\omega} \text{Im}[\chi(\omega)].$$

4. **Practical Applications:** We provided formulas for peak frequencies, quality factors, and parameter estimation from spectral measurements.

These results provide a complete frequency-domain characterization of the stochastic damped harmonic oscillator, essential for experimental spectroscopy, noise analysis, and signal processing applications.

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A Derivation of the Power Spectrum via Fourier Transform

For the underdamped case, the position autocorrelation is:

$$C_{xx}(\tau) = \frac{\sigma^2}{2ab} e^{-\alpha|\tau|} \left[\cos(\omega|\tau|) + \frac{\alpha}{\omega} \sin(\omega|\tau|) \right].$$

The Fourier transform is:

$$S_{xx}(\Omega) = \int_{-\infty}^{\infty} C_{xx}(\tau) e^{-i\Omega\tau} d\tau.$$

$$= \frac{\sigma^2}{ab} \int_0^\infty e^{-\alpha\tau} \left[\cos(\omega\tau) + \frac{\alpha}{\omega} \sin(\omega\tau) \right] \cos(\Omega\tau) d\tau.$$

Using the standard integrals:

$$\int_0^\infty e^{-\alpha\tau} \cos(\omega\tau) \cos(\Omega\tau) d\tau = \frac{\alpha(\alpha^2 + \omega^2 + \Omega^2)}{2[(\alpha^2 + \omega^2 + \Omega^2)^2 - 4\omega^2\Omega^2]},$$

$$\int_0^\infty e^{-\alpha\tau} \sin(\omega\tau) \cos(\Omega\tau) d\tau = \frac{\omega(\alpha^2 + \omega^2 - \Omega^2)}{2[(\alpha^2 + \omega^2 + \Omega^2)^2 - 4\omega^2\Omega^2]}.$$

Substituting:

$$S_{xx}(\Omega) = \frac{\sigma^2}{ab} \left[\frac{\alpha(\alpha^2 + \omega^2 + \Omega^2)}{2\Delta} + \frac{\alpha}{\omega} \frac{\omega(\alpha^2 + \omega^2 - \Omega^2)}{2\Delta} \right],$$

where $\Delta = (\alpha^2 + \omega^2 + \Omega^2)^2 - 4\omega^2\Omega^2$.

$$= \frac{\sigma^2}{ab} \left[\frac{\alpha(2\alpha^2 + 2\omega^2)}{2\Delta} \right] = \frac{\sigma^2\alpha(\alpha^2 + \omega^2)}{ab\Delta}.$$

Since $b = \alpha^2 + \omega^2$:

$$S_{xx}(\Omega) = \frac{\sigma^2\alpha}{a\Delta}.$$

But $a = 2\alpha$, so:

$$S_{xx}(\Omega) = \frac{\sigma^2}{2\Delta}.$$

Now $\Delta = (\alpha^2 + \omega^2 + \Omega^2)^2 - 4\omega^2\Omega^2$.

$$= (\Omega^2 + b)^2 - 4\omega^2\Omega^2 = \Omega^4 + 2b\Omega^2 + b^2 - 4\omega^2\Omega^2 = \Omega^4 + (2\alpha^2 + 2\omega^2 - 4\omega^2)\Omega^2 + b^2.$$

$$= \Omega^4 + (2\alpha^2 - 2\omega^2)\Omega^2 + b^2.$$

Since $a = 2\alpha$ and $b = \alpha^2 + \omega^2$:

$$\Delta = (b - \Omega^2)^2 + a^2\Omega^2.$$

Therefore:

$$S_{xx}(\Omega) = \frac{\sigma^2}{(b - \Omega^2)^2 + a^2\Omega^2}.$$

□

B Fourier Transform Table

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Function	Fourier Transform
$e^{-\alpha \tau }$	$\frac{2\alpha}{\alpha^2 + \omega^2}$
$e^{-\alpha \tau } \cos(\omega_0 \tau)$	$\frac{\alpha}{(\alpha^2 + (\omega - \omega_0)^2)} + \frac{\alpha}{(\alpha^2 + (\omega + \omega_0)^2)}$
$e^{-\alpha \tau } \sin(\omega_0 \tau)$	$\frac{\omega_0}{(\alpha^2 + (\omega - \omega_0)^2)} - \frac{\omega_0}{(\alpha^2 + (\omega + \omega_0)^2)}$
$e^{-\alpha \tau } \text{sgn}(\tau) \sin(\omega_0 \tau)$	$-\frac{2i\alpha\omega_0}{(\alpha^2 + \omega^2 + \omega_0^2)^2 - 4\omega^2\omega_0^2}$

Table 2: Fourier transform table for common functions.

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